



FOSSEE 2026 - SEMESTER LONG INTERNSHIP

INTERNSHIP SCREENING TASK REPORT

Task 7 — PCB Design using eSim KICAD

Prepared by Kadarkarai V, II Year

**Department of VLSI Design and Technology
Chennai Institute of Technology**

PROJECT: STM32BASE — Custom STM32F103C8T6 USB Development and Multi-Interface Controller Board

1. Project Overview

This project presents the complete printed circuit board (PCB) design and schematic files for a custom STM32F103C8T6 microcontroller development and interface board, named STM32CORE. The design was developed as part of the screening task for the FOSSEE Autumn Internship 2026 program conducted by IIT Bombay.

The primary objective of this project was to design a compact, functional, and manufacturable controller board centered around the STM32F103C8T6 (Cortex-M3) microcontroller, integrating regulated power, USB connectivity, UART communication, I²C expansion, SWD programming, boot-mode selection, and an external crystal clock reference onto a single custom two-layer PCB.

Key Project Details

Category	Specification / Detail
Microcontroller	STM32F103C8T6 / STM32F103CBTx (ARM Cortex-M3, up to 72 MHz)
Design Tools	eSim 5 (schematic capture), KiCad 6 (PCB layout, integrated via eSim)
Input Supply	5 V from USB (VBUS)
Regulated Supply	3.3 V, via AMS1117-3.3 LDO regulator
Crystal Oscillator	16 MHz external crystal
PCB Stackup	2-layer (Front Copper + Back Copper)
Verification Status	Both ERC and DRC passed with zero errors

The finished design includes a fully routed two-layer PCB, verified schematic with zero ERC violations, a zero-violation DRC report, and a 3D rendered model for visual verification of component placement and mechanical fit.

2. Circuit Description

2.1 Power Supply Section

The board is powered from the 5 V USB VBUS rail and regulated down to a stable 3.3 V supply using an AMS1117-3.3 LDO regulator (U1). Bulk input and output capacitors (C1, C2 — 22 μ F each) stabilise the supply on both sides of the regulator, and a status LED (D1) with a 1.5 k Ω current-limiting resistor (R1) provides visual power-on indication.

2.2 Clock and Reset

A 16 MHz external crystal (Y1) with a pair of 10 pF load capacitors (C12, C13) provides the primary system clock reference to the STM32F103 oscillator pins (PD0/PD1). The MCU's internal PLL derives the required core clock from this external source. A dedicated reset line (NRST) and BOOT0 configuration network (R2, SW1) allow selection between normal firmware execution and system bootloader mode for programming/recovery.

2.3 Connectivity and Interfaces

- Micro-USB Port (J1): USB 2.0 Full-Speed interface for power and USB D+/D– data, with a 1.5 k Ω series resistor (R3) on the data line as per USB application guidelines.
- SWD Header (J3): 4-pin header exposing SWDIO, SWCLK, 3.3 V and GND for direct on-board programming and debugging.
- UART Header (J2): 4-pin header exposing USART_TX, USART_RX, 3.3 V and GND for serial communication and debugging.
- I²C Header (J4): 4-pin header exposing I2C2_SDA, I2C2_SCL, 3.3 V and GND, with 1.5 k Ω pull-up resistors (R4, R5) for sensor and peripheral expansion.

2.4 Power Filtering and Decoupling

The analog supply (VDDA) is isolated from the digital 3.3 V rail using a 120 Ω ferrite bead (FB1) with local filter capacitors (C8, C9, C10), reducing high-frequency supply noise reaching the MCU's analog section. Digital decoupling is provided at the MCU supply pins using a bank of ceramic capacitors (C3–C7: 10 μ F and 100 nF values) placed close to the VDD/VBAT pins.

2.5 Peripheral Summary

Interface	Header / Pins	Typical Application
USB	Micro-USB (J1), USB D+/D–	PC communication, firmware/programming tools
UART	J2 — USART_TX, USART_RX	Debugging, GPS, GSM, Bluetooth modules
I ² C	J4 — SDA, SCL	Sensors, displays, EEPROMs
SWD	J3 — SWDIO, SWCLK	STM32 programmer / debugger
BOOT	SW1 (SPDT) + R2	Firmware boot-mode / recovery selection

3. Schematic Screenshots

The complete schematic was designed in eSim 5 and organised into two logical, clearly labelled functional blocks — Power Supply and Microcontroller with USB — for clarity and maintainability.

3.1 Full Schematic Overview

The top-level schematic sheet (STM32BASE_DESIGN) provides a complete view of the design, showing the Power Supply block and the Microcontroller-with-USB block interconnected through labelled nets and power buses, along with the board mounting-hole reference symbols.

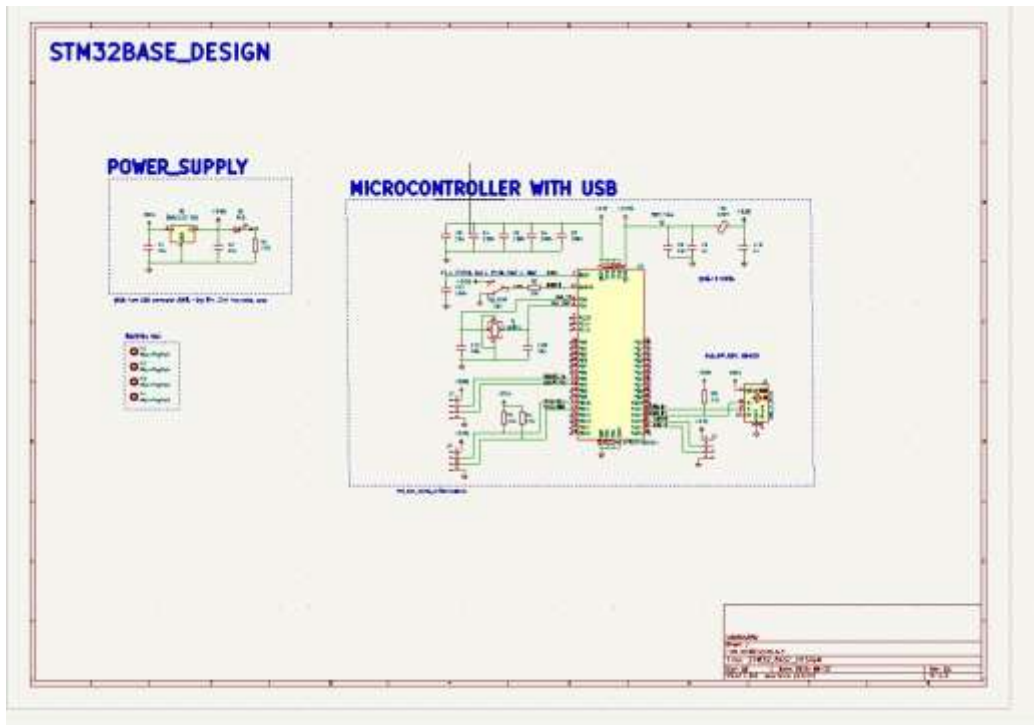


Figure 1: Full schematic overview — Power Supply and Microcontroller with USB blocks.

3.2 Schematic Section 1 — Power Supply

This section shows the AM1117-3.3 (AMS1117-3.3) regulator circuit that converts the 5 V USB VBUS supply into a regulated 3.3 V rail, along with the input/output bulk capacitors, the power-indication LED and its current-limiting resistor, and the four PCB mounting-hole references.

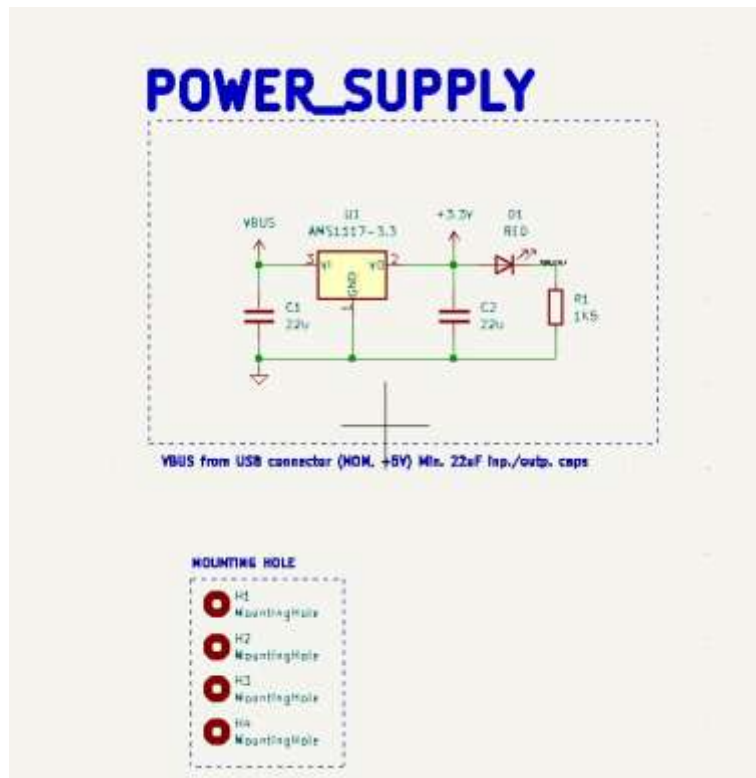


Figure 2: Power Supply schematic — AMS1117-3.3 regulation and mounting-hole references.

3.3 Schematic Section 2 — Microcontroller with USB

This section captures the STM32F103CBTx pin assignments, the full decoupling capacitor network, the 16 MHz crystal oscillator connections, NRST/BOOT0 configuration circuitry, the VDDA filtering network, and the USB, UART and I²C interface connections — all cross-referenced against the STM32CubeIDE pin-out.

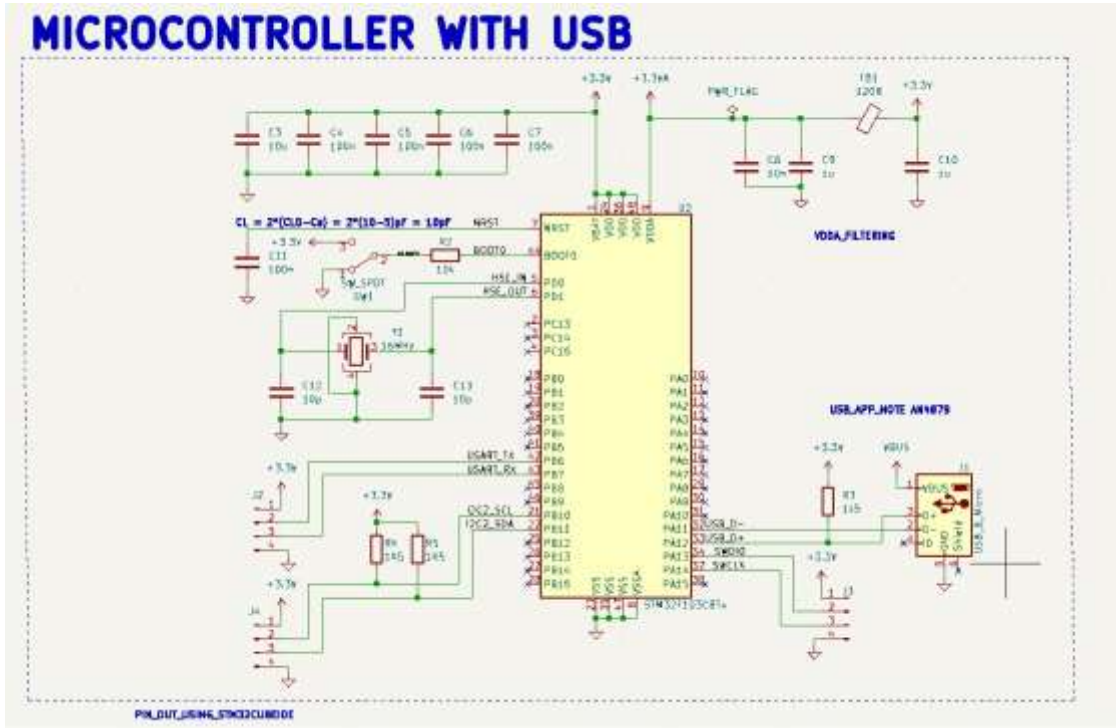


Figure 3: Microcontroller with USB — MCU core, clock, boot, and interface circuitry.

3.4 Electrical Rules Check (ERC)

The Electrical Rules Checker was run on the complete schematic in eSim/KiCad and returned zero violations, confirming that all nets, power pins, and interface connections are electrically consistent.

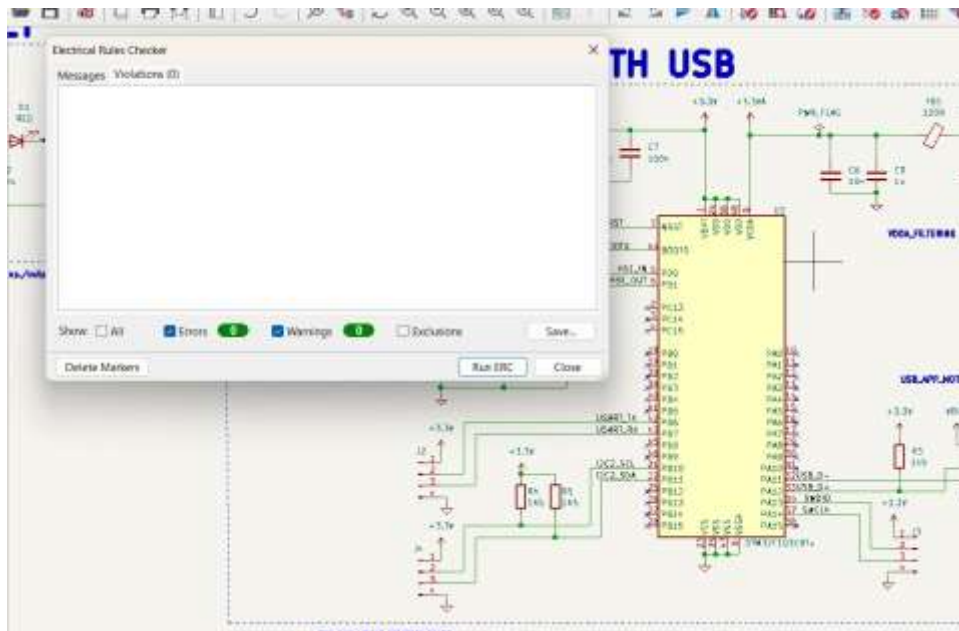


Figure 4: Electrical Rules Checker — 0 errors, 0 warnings.

4. Footprint Assignment Details

Component footprints were assigned in KiCad 6 using standard, datasheet-verified footprint libraries appropriate to each package type, ensuring correct pad geometry and courtyard clearances for fabrication.

4.1 Microcontroller

- Component: STM32F103C8T6 / STM32F103CBTx
- Package: LQFP-48, 7×7 mm body, 0.5 mm pin pitch

4.2 Passive Components

- Resistors: 0603 SMD footprint — used for pull-ups (I²C), current limiting (LED, BOOT0), and USB series termination
- Capacitors: 0603 SMD footprint — used for decoupling, bulk bypass, VDDA filtering, and crystal load capacitors
- Crystal Oscillator: SMD 2-pin/4-pin package as per component datasheet (16 MHz)

4.3 Connectors

- Micro-USB: Standard Micro-B USB receptacle footprint with mechanical shield/anchor pads
- Pin Headers (UART, SWD, I²C): 1×4, 2.54 mm pitch through-hole male headers

4.4 Power and Interface Components

- Voltage Regulator (AMS1117-3.3): SOT-223 footprint with thermal pad
- Boot Selector: SPDT slide/tactile switch footprint
- Status LED: 0603 SMD LED footprint

5. PCB Layout

The PCB layout was completed in KiCad 6 (accessed through eSim 5 integration) as a two-layer board, with the Front Copper (F.Cu) layer carrying the primary component-side routing and the Back Copper (B.Cu) layer used for return paths and selective routing.

5.1 Complete Board — Routed Layout

The routed board places the STM32F103 MCU (U2) centrally, with the crystal and its load capacitors kept close to the oscillator pins, the USB connector positioned at the board edge, and the UART, SWD and I²C headers broken out along the top and bottom edges for easy access. Four mechanical mounting holes are provided at the corners.

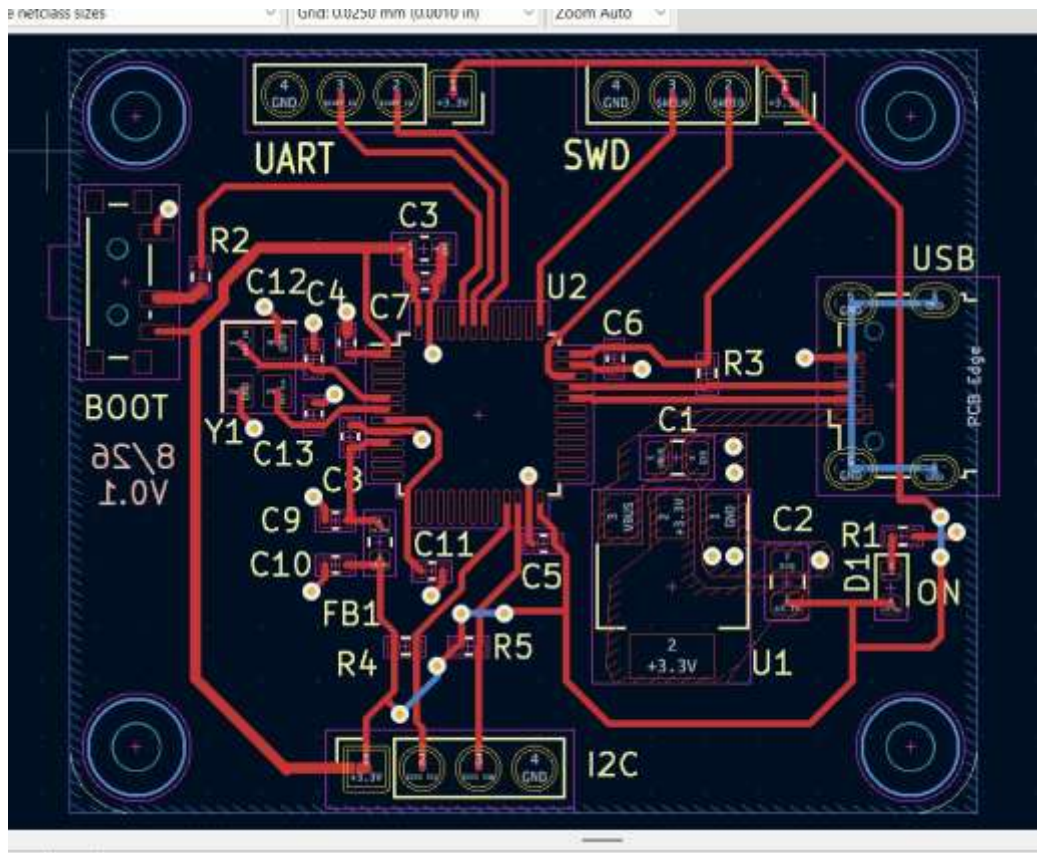


Figure 5: Complete routed PCB layout — top view showing UART, SWD, USB, I²C and BOOT sections.

5.2 Board Layout — Alternate Layer View

This view highlights the copper routing on the board with the alternate KiCad colour theme, showing the same interconnects between the MCU, power section (U1, C1, C2), filtering network (FB1, R4, R5) and the four peripheral headers.

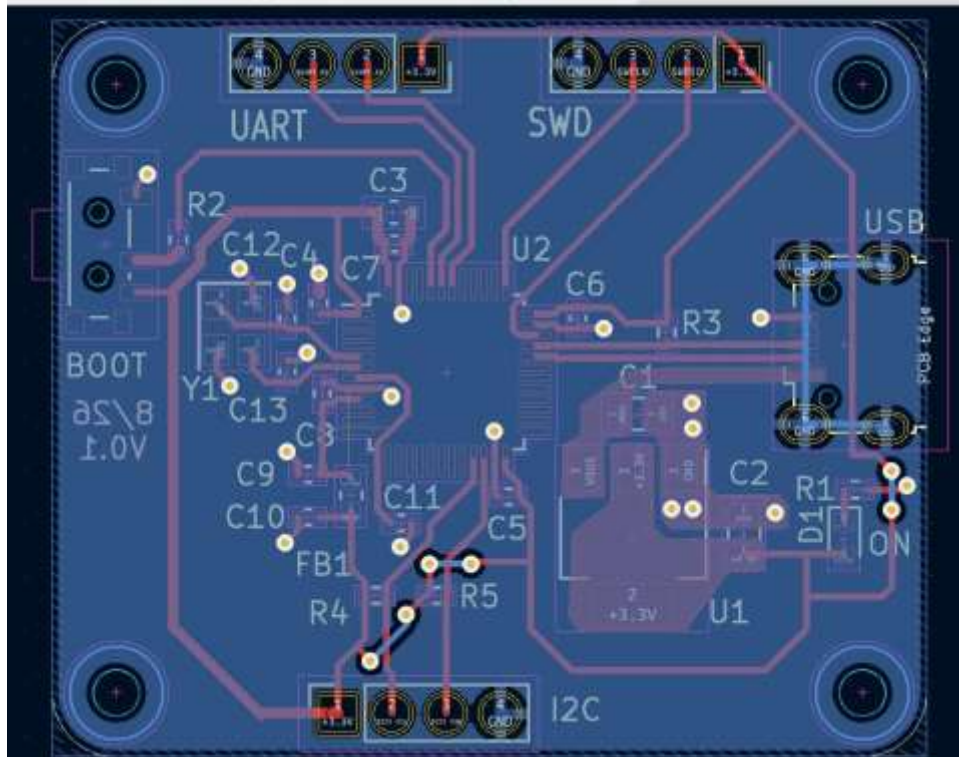


Figure 6: PCB routing — alternate layer colour view.

5.3 Detailed Routing View

A closer view of the board confirms the routing quality around the MCU, decoupling network and connector regions, with short, direct traces maintained for the crystal and USB data lines.

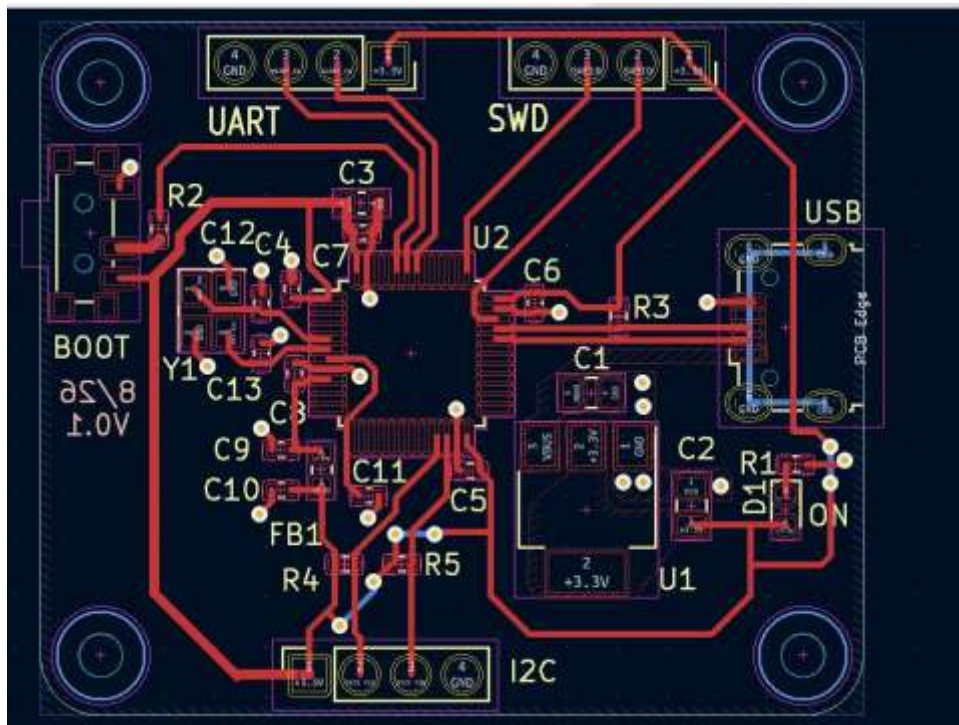


Figure 7: Detailed PCB routing view around the MCU and power sections.

6. Routing Methodology

6.1 Layer Stackup

Layer	Name	Function
1	F.Cu (Front)	Primary component-side signal routing
2	B.Cu (Back)	Secondary routing and return paths

6.2 Signal Routing Guidelines

- Decoupling capacitors placed as close as physically possible to the MCU VDD/VBAT pins.
- Crystal oscillator traces routed short and direct, with the crystal and load capacitors kept close to the MCU oscillator pins to minimise parasitic capacitance.
- USB D+/D– lines routed as a short, symmetric pair from the MCU to the Micro-USB connector at the board edge.
- I²C pull-up resistors located directly at the I²C header for signal integrity.
- General signal traces routed on the front copper layer with return paths managed on the back copper layer.
- Components kept clear of the four mounting-hole regions for mechanical safety.

6.3 Design Rule Constraints

- Minimum trace width and spacing as per standard two-layer manufacturing capability.
- Minimum via drill size and annular ring maintained within manufacturer tolerances.
- Copper-to-edge clearance maintained around the full board outline.
- No acid traps or acute-angle traces present in the routing.

7. DRC Results

The Design Rules Check was performed on the completed two-layer PCB layout in KiCad 6 to verify physical manufacturability. All copper zones were refilled before running the check.

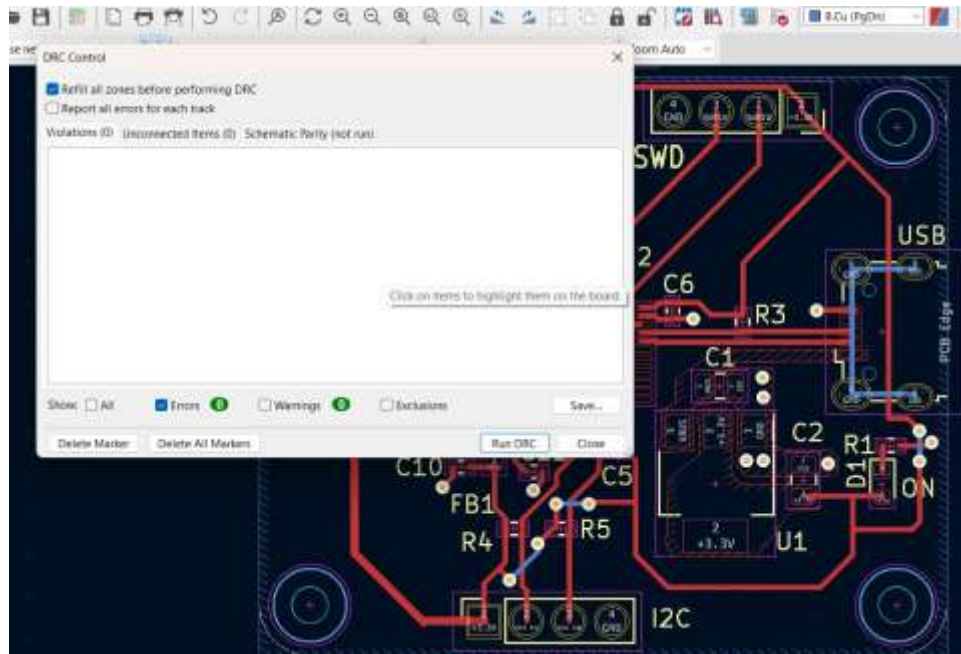


Figure 8: DRC Control — 0 violations, 0 unconnected items.

The DRC report confirms zero violations across errors, warnings and unconnected items, indicating that all traces, vias, clearances, and drill holes are within standard manufacturing capability.

8. 3D Model

The 3D model of the board was generated using KiCad's built-in 3D viewer to verify component placement accuracy, connector accessibility, and overall board form factor before committing to fabrication.

8.1 Front View

The front (top) view shows the primary component side of the board, with the STM32 MCU (U2) at the centre, surrounded by the crystal (Y1), decoupling capacitors, the BOOT switch, the regulator section (U1), and the UART, SWD, USB and I²C connectors clearly labelled on the silkscreen.

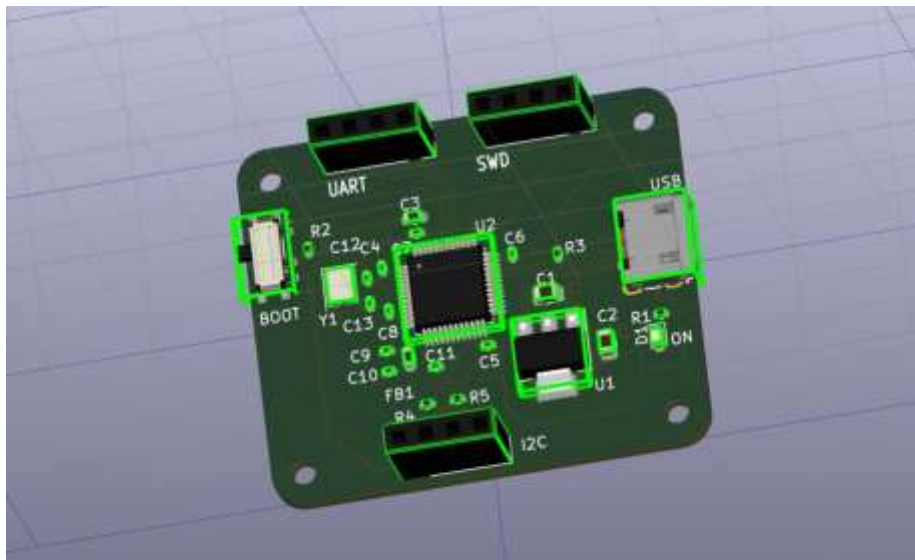


Figure 9: 3D model — front view with labelled interfaces.

8.2 Back View

The back (bottom) view shows the underside of the board, including the header through-hole pads, the version/date silkscreen marking (8/26, V0.1), and the mounting-hole positions.

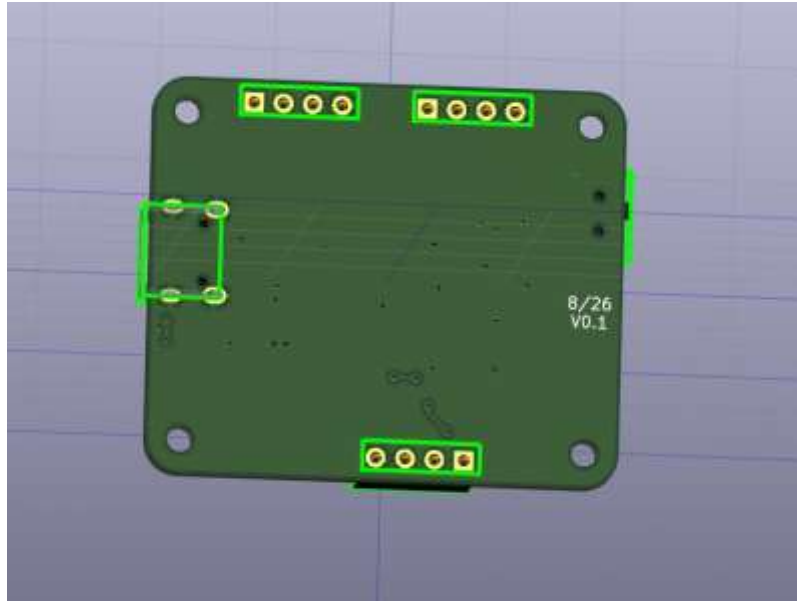


Figure 10: 3D model — back view.

9. Design Considerations and Conclusion

9.1 Design Considerations

Power Integrity: Bulk capacitors at the regulator input/output and a dedicated decoupling bank at the MCU supply pins keep the 3.3 V rail stable under load.

Signal Integrity: Short, direct routing for the crystal and USB data lines minimises parasitic effects and preserves clock accuracy and USB signal quality on a two-layer stackup.

Analog Isolation: The VDDA filtering network (FB1 ferrite bead with local capacitors) isolates the MCU's analog supply from digital switching noise.

Manufacturability: The design satisfies the FOSSEE eSim PCB Design requirement of a minimum of two copper layers, and passes DRC with zero errors, confirming that all features are within standard manufacturing capability.

Expandability: Unused STM32F103 peripherals (SPI, additional ADC/timer channels, CAN) remain available for a future, more advanced revision of the board.

9.2 Conclusion

This project successfully delivers a complete, verified PCB design for the STM32CORE development and interface board built around the STM32F103C8T6 microcontroller. The design achieves the following key outcomes:

- Fully verified design — ERC passed with zero violations, and DRC passed with zero errors and zero unconnected items.
- Compact two-layer stackup — meeting the FOSSEE eSim PCB Design minimum layer requirement while remaining realistic to fabricate.
- Comprehensive connectivity — the board exposes USB, UART, I²C and SWD through accessible headers, along with BOOT-mode selection.
- Complete documentation — schematic, PCB layout, ERC/DRC verification and 3D visualisation are all included for full design transparency.

The design demonstrates proficiency in schematic capture, PCB layout and routing, design-rule compliance, and 3D visualisation using eSim 5 and KiCad 6 — the core competencies required for the FOSSEE Autumn Internship 2026 screening task.

Designed by Kadarkarai V | Department of VLSI Design and Technology, Chennai Institute of Technology | FOSSEE Autumn Internship 2026 — IIT Bombay